label-guard: an OpenCV 5 perception–decision–action loop that knows when not to answer

Technical report · OpenCV AI Competition 2026, powered by AWS · Focus path: **Agentic Vision with OpenCV 5** · solo entrant · live demo (no login): https://hqadhnbyzbv5dyok3zdnnfpcae0tcozx.lambda-url.ap-northeast-2.on.aws/

Problem. A person with a food allergy, or the family member reading for them, photographs the ingredient list on the back of a package. A phone photo is often blurred, glared or tilted, and a tool that reads a bad photo can still say "no milk found" — a silent miss.

What we built. The competition asks that "image or video results must influence a subsequent plan, tool call, action, or request for human approval" (opencv26.devpost.com, "Agentic Vision with OpenCV 5"). In label-guard, OpenCV 5 first judges whether the photo is good enough to read (perspective rectification, blur, glare, text regions). That result decides the next step: **retake request** (the reader is not called), or read with the OpenCV 5 DNN text reader and then **allergy warning**, **human check** when the read is uncertain, or **no warning** — always with "final check by a person". It runs on AWS Lambda behind a public Function URL.

Key numbers (measured). On 104 public label photos with the deployed reader: warning recall 0.87, precision 0.80, "safe recall" (warned or abstained) 0.91, 5 silent misses out of 55 positive photos. On 1,248 synthetic retake pairs: verdict change rate 0.38, abstention 0.38 → 0.10 after retake, useful abstention only 0.26. Live endpoint: 30/30 HTTP 200, round trip p50 3.36 s.

Re-measured with the deployed reader (OpenCV 5 DNN). The same 1,248 retake pairs give verdict change 0.38 and abstention 0.38 (after retake 0.10), the same shape as with the macOS reader, but synthetic-degradation silent misses rise from 54 to 83. With the quality gate on / off, silent misses over real photos are 20 / 20 (group × photo) and recall 0.74 / 0.76, with fish at 0.33 and egg at 0.56. So the gate's measured effect is limited to blur and low resolution; glare and tilt remain weak. Synthetic degradations plus 42 real pairs of different uploads (agreement 0.76); no same-person phone retakes.

> **What only label-guard has, among the tools below:** it measures whether *this photo* is good enough to rule an allergen out, and when it is not, it refuses to say "none" and changes its next action (retake or human check) — and it publishes how often that abstention is right (0.26) and how often it still misses silently (83 of 1,248 degraded photos; per-ingredient recall in 6.1).

Existing tools for the same job (public-page basis, 2026-10-26)

Tool	How it works	What it does not do (vs. label-guard)
Fig https://foodisgood.com/	Scan a barcode; ingredients come from the app's product database and are matched to diet/allergy filters (site: "scan any barcode … see how any ingredient matches").	Per its public page, input is a barcode matched to its database; we did not test whether it also judges a label photo.
Yuka https://yuka.io/en/	Scans products and scores them from a product database (site: "scans and analyzes labels in the blink of an eye").	Per its public page, a product score from a database; we did not use the app.
Microsoft Seeing AI https://apps.apple.com/us/app/seeing-ai/id999062298	Reads text aloud; "Document alignment provides audio cues to capture a printed page".	Guides framing, then reads; its public pages do not describe a decision on whether an allergen can be ruled out.
General VLM apps (ChatGPT / Gemini photo Q&A) https://chatgpt.com/ · https://gemini.google.com/	Free-form answers about an uploaded photo; a user can ask "does this label contain milk?" (public-page basis).	Public pages do not describe a photo-quality check or a measured abstention rate; we did not test them hands-on.
Be My AI (Be My Eyes) https://www.bemyeyes.com/be-my-ai/	Free-form AI description of a photo (site: "From reading labels to exploring new places").	Answers in prose; no measured abstention when the photo is too poor to answer.

The four app rows are based on each tool's public page only (no hands-on use); the general VLM row describes typical photo Q&A behaviour on public product pages and was not tested hands-on either. We do not claim higher reading accuracy than these products; we did not benchmark them. We do not compete on real-time video or language-model agents; the claim is limited to **measured abstention**. The difference claimed is the loop: photo quality and reading confidence change the action, and the abstentions are measured.

Report structure (follows the rules' list, in the same order)

1. Problem · 2. Users · 3. Architecture · 4. OpenCV 5 implementation · 5. AWS deployment · 6. Evaluation · 7. Limitations · 8. Responsible use

1 Problem

Allergen information is printed as free text on the package. Reading it from a phone photo has two failure modes that matter differently: a **false alarm** costs a second look, while a **silent miss** ("no warning" on a product that contains the allergen) can cost a reaction. Most failures start before any text recognition: the photo is blurred, a glossy film reflects light over the words, or the label is shot at an angle.

So the question we answer is not only "what does the label say?" but "**is this photo good enough to say that an allergen is absent?**" If it is not, the right output is an action — take the photo again, or ask a person to read the original — not a guess.

2 Users

- **People with food allergies** checking a product in a shop or at home, including imported products whose barcodes may not be in a product database.
- **Family members and carers** who read labels for a child or an older relative, often from a photo sent by message.
- **Low-vision users**: the page works with the keyboard only, results are announced through an `aria-live` region, and every verdict is plain text (Section 6.5).

Use is one photo → one verdict with its reason, in a few seconds. The live page is in English by default (`?lang=ko` for Korean); the evaluation photos are Korean, English, French and Spanish labels. We have **not** run a study with real users, so we make no claim about time saved or reactions prevented.

3 Architecture

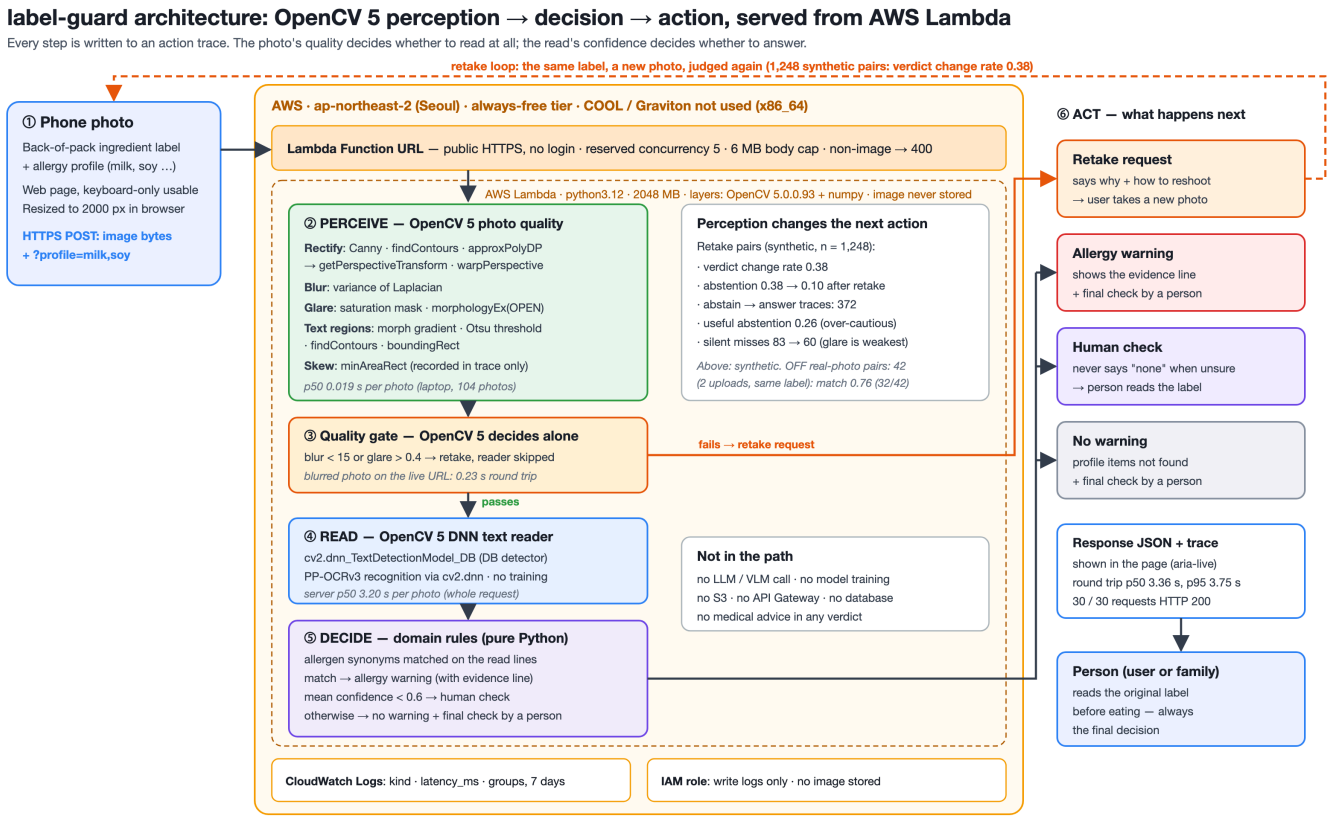


Figure 1. Components and data flow. ① phone photo → Lambda Function URL → ② OpenCV 5 photo quality → ③ quality gate (OpenCV 5 alone can stop here with a retake request) → ④ OpenCV 5 DNN reader → ⑤ domain rules → ⑥ one of four actions, returned with the action trace. COOL and AWS Graviton are not used.

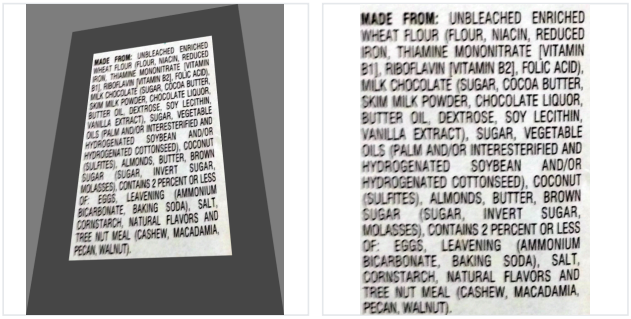
Code layout (Clean Architecture). `domain/` (verdict rules, allergen synonyms; no OpenCV, AWS or file I/O) ← `application/` (ports: `QualityAnalyzer`, `TextReader`; the judge use case) ← `adapters/` (`opencv5_quality`, `dnn_reader` for Linux/AWS, `vision_reader` for macOS, evaluation runners) ← `infrastructure/` (`lambda_handler`). Swapping the reader does not touch the decision rules; both readers were evaluated with the same rules and thresholds.

The loop. The "agent" here is a deterministic policy, not a language model. Its perception is OpenCV 5; its decision is explicit rules; its actions are retake, warn, ask a person, or report no warning. A retake request names the reason and how to fix it (blur: hold steady 10–15 cm away; glare: tilt away from the light; framing: shoot straight on), then the new photo of the same label which is judged again — the retake pairs in Section 6.3 measure how often that changes the verdict. A human check is the system's request for human approval: it is returned whenever the reader's mean confidence is below 0.6, instead of "no warning".

4 OpenCV 5 implementation

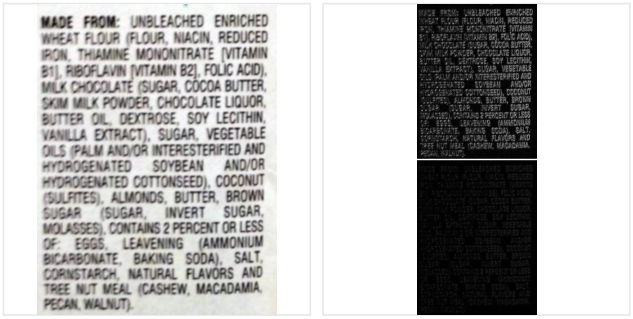
All vision runs in `opencv-python-headless==5.0.0.93` (`cv2.__version__ = 5.0.0`). No model is trained; the DNN reader uses public pre-trained ONNX weights loaded with `cv2.dnn`. Thresholds: blur score < 15 or glare ratio > 0.4 → retake; mean read confidence < 0.6 → human check; the text-region minimum is set to 0 (off) because the classical detector found fewer lines than the reader (Section 6.1).

1. Perspective rectification



`Canny` → `findContours` → `approxPolyDP` → `getPerspectiveTransform` + `warpPerspective`. Synthetic tilt (left): text regions 0 before, 58 after. The warped image is kept only if it has at least as many text regions.

2. Blur score



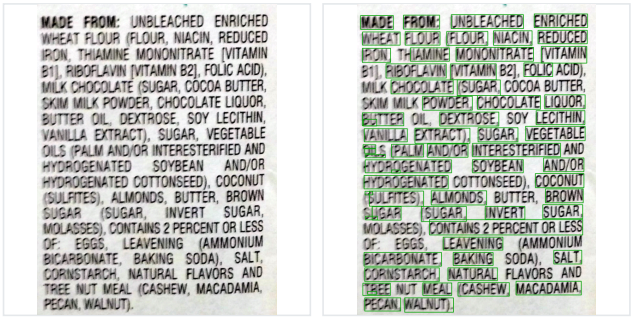
Variance of `Laplacian` on a 1000 px wide grey image. Original 274.5, synthetic blur (left) 11.5 < 15 → retake. Right: |Laplacian| of original (top) and blurred (bottom).

3. Glare ratio



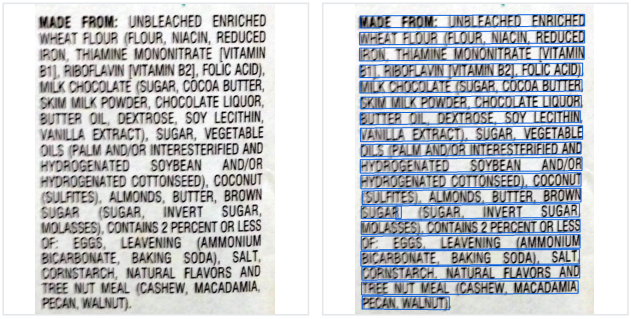
Pixels saturated in all channels, then `morphologyEx(MORPH_OPEN)` with a 15 px ellipse keeps only blobs (red). Original 0.006, synthetic glare 0.309 — still under 0.4, so this photo is *not* sent back: glare is our weakest gate (Section 6.3).

4. Text regions



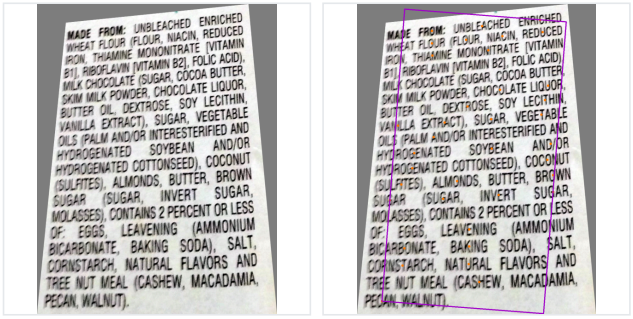
`morphologyEx(MORPH_GRADIENT)` → `threshold(OTSU)` → `morphologyEx(MORPH_CLOSE)` → `findContours` + `boundingRect` : 64 line candidates. Used to accept or reject rectification and logged in the trace.

5. DNN text detection and recognition



`dnn_TextDetectionModel_DB` finds 20 text polygons; each is unwarped with `warpPerspective` , split into words by an Otsu ink profile, and read by PP-OCRv3 through `dnn.readNet` + `blobFromImage` . Example line: “WHEAT FLOUR (FLOUR, NIACIN, REDUCED” (confidence 0.96).

6. Skew (trace only)



`minAreaRect` over text-line centres. It is written to the trace but not used in any decision: on tall text blocks the estimate is unreliable (89.0° on the untilted original).

Figure 2. Each operation’s input (left) and output (right) on public photo 0014100098423 (Open Food Facts, CC BY-SA). Tilt, blur and glare inputs are synthetic degradations of that photo. Regenerate with `scripts/make_report_figures.py` .

How much OpenCV decides. The classical quality gate alone decided 6.7% of verdicts (7/104, all retake requests); these photos never reach the reader. Every other verdict needs the read lines. On AWS the reader itself is OpenCV (`cv2.dnn`), so all image processing is OpenCV 5; matching and the decision are plain Python rules, and no LLM or VLM is called. The trace tags the read step `[OpenCV5-DNN]` and the rules `[rules]` . Time per photo: OpenCV quality steps 6% of processing, reading the rest. Rectification was applied to 3/104 evaluation photos, because most are already cropped. Unit tests: 79 passing (`pytest -q`).

5 AWS deployment

Service	Role	Cost
Lambda (python3.12, x86_64, 2048 MB, 60 s timeout)	Runs quality analysis, DNN reader and decision	Always-free: 1M requests and 400,000 GB-s per month
Lambda Function URL (AuthType NONE)	Public HTTPS entry; `GET /` upload page, `POST /?profile=milk,soy` image → JSON	No extra charge
2 Lambda layers	OpenCV 5 (31 MB) and shared libraries + numpy (46 MB), split to fit the 50 MB direct-upload limit (no S3, no ECR)	No charge
CloudWatch Logs (7-day retention)	One line per request: verdict kind, latency ms, number of groups; Lambda REPORT lines	Always-free 5 GB
IAM role	`AWSLambdaBasicExecutionRole` only — no S3, database or network permissions; the function holds no secrets	No charge

Reliability and abuse limits. Reserved concurrency 5 (excess requests get 429), 6 MB body cap (413), non-image bodies 400. Uploaded images are decoded in memory and discarded; no image is written to storage or logs. **Cost.** One request ≈ 3 s × 2 GB = 6 GB-s, so about 66,000 requests a month stay inside the always-free compute; 1,000 judging requests use 1.5% of it. **Reproduce.** `./scripts/build_lambda.sh` builds the three zips; `python3 scripts/deploy_lambda.py <profile> ap-northeast-2` publishes layers, creates or updates the function and prints the URL (idempotent). Local tests: three README commands.

Availability. We plan to keep the endpoint online at least through the OpenCV Live! presentation on 2026-11-10, inside the always-free tier. Concurrency is capped at 5 simultaneous requests; excess requests receive HTTP 429 and the page shows an English “demo is busy, try again in a few seconds” message. No budget alarm is set (expected use stays near 1.5% of the free compute), so a burst of judges is limited by the cap, not by cost.

Measured on the public URL (2026-10-26, 30 requests, 3 photos × 10, about 260 KB JPEG each): 30/30 HTTP 200; client round trip p50 3.36 s, p95 3.75 s; server processing p50 3.20 s, p95 3.52 s; one cold start 5.77 s round trip; max memory 377 MB (targets set in advance: p95 ≤ 60 s, server p50 ≤ 10 s). A blurred photo returns a retake request in 0.2–0.4 s (0.23 s in the 30-request test; 0.41, 0.2 and 0.22 s in the three traces of 6.2) because the reader is skipped.

6 Evaluation

Data. 104 ingredient photos from Open Food Facts (Korean, English, French, Spanish), with crowd-sourced allergen tags as ground truth; 0 photos failed to run. Groups: milk, gluten, egg, soy, nut, fish, other. **Metrics.** *Recall* = warned positives / positives. *Safe recall* = (warned + retake or human check) / positives — the share where the system did not say "none". *Silent miss* = positive photo answered "no warning" (the most serious error). *Verdict change rate* = retake pairs whose verdict kind differs. *Useful abstention* = abstentions where forcing an answer would have been wrong.

6.1 Photo-level allergen detection (profile = all groups, 55 positive photos)

Reader (same rules, same thresholds)	Precision	Recall	Safe recall	Silent misses	Read time / photo (M-series laptop)
OpenCV 5 DNN (deployed on AWS)	0.80	0.87	0.91	5	0.40 s
OpenCV 5 DNN, held-out 52 photos (28 positive; thresholds chosen on the other 52)	0.83	0.86	0.93	2	—
macOS Vision (on-device, comparison)	0.79	0.89	0.93	4	0.59 s

Held-out check. We split the 104 photos with a fixed seed into 52 for tuning and 52 held out, re-picked the thresholds on the tuning half only (grid of 125 combinations, rule fixed before looking: maximum photo-level recall with precision ≥ 0.75), and scored the held-out half: with the deployed values (blur 15, glare 0.4, confidence 0.6) the held-out half gives photo-level recall 0.86 (24/28) against 0.89 (24/27) on the tuning half, 2 silent photos; per group × photo recall 0.75 (30/40) with 8 silent misses. The tuning half picked blur 15 and glare 0.4 again; it preferred confidence 0.8 over the deployed 0.6, which on the held-out half would cut the group-level silent misses from 8 to 6 at the same recall. We did not redeploy with 0.8 (source: `docs/eval/holdout.md`). The two readers give the same verdict kind on 94% of photos (98/104). Turning the quality gate off raises DNN recall to 0.89: on these already-cropped photos the gate costs one warning, which is the price of refusing to read bad photos. Per group × photo (micro average, DNN reader): precision 0.59, recall 0.74, safe recall 0.78, 20 silent misses; weakest groups are fish (recall 0.33) and egg (0.56).

Ingredient group (group × photo, DNN)	Positives	Recall, gate on	Recall, gate off	Silent misses on / off
Milk	30	0.97	0.97	1 / 1
Wheat/gluten	21	0.76	0.76	5 / 5
Egg	9	0.56	0.56	4 / 4
Soy	7	0.71	0.71	1 / 1
Peanut/nut	7	0.57	0.71	1 / 1
Fish	6	0.33	0.33	4 / 4
Other	10	0.60	0.60	4 / 4
All	90	0.74	0.76	20 / 20

The headline 0.87 is photo-level; per ingredient it is 0.74, and the gate on/off difference is one warning. The gate does not reduce silent misses on these already-cropped photos (source: `docs/eval/retake-linux.md`).

The target we set before measuring was photo-level recall ≥ 0.90 or a table of reasons. We reached 0.87; the reasons are in 6.4.

6.2 Live traces: a retake changes the action (public URL, 2026-10-26)

Product (Open Food Facts)	Blurred photo → verdict	Round trip	Same label, sharp photo → verdict	Round trip
0014100098423	retake_request (blur)	0.41 s	allergy_warning (gluten, milk, soy)	3.58 s
8801115381518	retake_request (blur)	0.2 s	allergy_warning (milk)	1.62 s
2935306002206	retake_request (blur)	0.22 s	allergy_warning (milk)	1.83 s

Profile milk, soy, gluten. The blurred photo is a synthetic degradation (Gaussian, level 2). In each pair the OpenCV 5 gate stops the first request before reading; the retake is read and warned with evidence lines. The demo video runs the first photo (0014100098423) with all seven groups ticked: the trace then lists five warned groups (gluten, milk, soy, nut, egg) instead of the three above; the photo and reader are the same, only the profile differs. Full JSON responses with traces are kept with this report.

6.3 Retake pairs and abstention (1,248 synthetic pairs)

Each of the 104 photos was degraded four ways (blur, glare, tilt, low resolution) at three strengths; the degraded photo plays "before retake" and the original "after retake". These are **synthetic** pairs, measured with the deployed OpenCV 5 DNN reader (macOS Vision cross-check: abstention 0.40 → 0.09, useful 0.28, silent misses 54 → 48); real photo pairs follow below; same-person phone retakes are planned, not measured.

Pairs	Verdict change	Abstention before → after	Abstain → answer	Useful abstention	Accuracy when answering, before → after	Silent misses before → after
1,248	0.38	0.38 → 0.10	372	0.26 (n=476)	0.77 (n=772) → 0.82	83 → 60

Degradation (each cell: verdict change / abstention / silent misses)	Strength 1	Strength 2	Strength 3
Blur	0.39 / 0.40 / 2	0.91 / 0.98 / 0	0.93 / 1.00 / 0
Glare	0.04 / 0.09 / 7	0.13 / 0.09 / 13	0.20 / 0.11 / 18
Tilt	0.09 / 0.08 / 9	0.10 / 0.07 / 11	0.13 / 0.10 / 12
Low resolution	0.29 / 0.31 / 6	0.57 / 0.59 / 5	0.73 / 0.78 / 0

Real photo pairs (not synthetic). 42 products, two different Open Food Facts uploads of one label each (CC BY-SA; 26 pairs by different uploaders, 16 by one). Not one person retaking.

Pairs	Verdict agrees	One side abstains	Abstention useful	Accuracy when answering
42	0.76 (32/42; 95% CI 0.61–0.87)	0.14 (6/42; always the worse photo)	1.00 (8 photos)	0.78 (59/76)

Of the 10 disagreeing pairs, 6 had one side handed to a person (safe) and 4 had two different answers (the risky case). Silent misses in the real pairs: 3 of 56 positive photos were answered “no warning” (one of them passed the blur gate at blur 24 and was then misread; the other two were sharp). The gate only changes who answers: on the same 76 answered photos accuracy is 0.78 with it on or off. Over all 84 photos, gate off gives 59 right and 25 wrong (0.70); gate on gives 59 right, 8 handed to a person and 17 wrong — the 8 handed over would otherwise have been answered wrongly. Mostly French labels, picked with a different OCR, so abstention is understated; no Korean pairs (source: `docs/eval/real-pairs.md`).

Reading the result honestly. Abstention works for blur: strong blur is always sent back and produces no silent miss. It is too cautious overall — only 0.26 of abstentions were needed — and it is too weak for glare: strong glare is caught in only 11% of photos and gives the most silent misses (18); tilt gives 12 at strength 3. The thresholds were not re-tuned on these degradations.

6.4 Failure cases (deployed DNN reader)

Cause counts over group × photo errors: reading errors 41, ingredient-word matching 26, low-confidence reading 2, glare 1.

Product	Expected	Verdict	Cause	Next fix
2000000021345 (ko)	nut present	no warning	reading error (388 of 502 characters read)	raise confidence floor; fuzzy word match
8801043060356 (ko)	fish present	no warning	ingredient-word matching	extend synonyms
8801161502646 (ko)	soy present	human check	low confidence (42 of 109 characters)	none — correct fallback
0014100098423 (en)	other present	no warning	reading error on the “other” group words	fuzzy word match
8437001415960 (es)	nut present	retake request	glare on the label	angle guidance; re-tune glare
8801043060356 (ko)	no milk / gluten / soy	warning	label line reads “contains wheat, soy, milk, beef”; the crowd tags omit them	ground-truth noise; not relabelled
8801037065626 (ko)	no gluten	warning	line reads “wheat flour (wheat: USA, Australia)”; tag missing	ground-truth noise; not relabelled

The last two rows show that some “false positives” are tags missing from the crowd-sourced truth. We kept the truth as published, so precision is likely understated; we did not correct it to avoid tuning the truth to the system.

6.5 Accessibility check (live page)

Keyboard-only upload and submit; labelled inputs; result in `role=status aria-live=polite`; focus moves to the result heading; 3 px focus outline; no meaning carried by colour alone. Not done: listening tests with a screen reader.

7 Limitations

- **Synthetic retakes.** All 1,248 retake pairs are synthetic degradations of public photos, measured with the deployed DNN reader. The only real photos are 42 pairs of different uploads of the same label (not one person retaking), mostly French labels; 0 real same-person phone retakes.
- **Tuned on the test photos.** The final thresholds (blur 15, glare 0.4) were lowered after seeing the 104 photos (fixed-in-advance thresholds 60 / 0.12 gave 24 retakes and recall 0.78), and the DNN reader’s word-gap ratio was changed once on the same photos (recall 0.855 → 0.873). Results may be optimistic; a 52/52 split check (6.1) held recall at 0.86, but 28 held-out positives is a small sample and the deployed values were first chosen on all 104 photos.
- **Abstention quality.** Useful abstention is 0.26: most abstentions were not needed. Glare detection misses strong glare most of the time; 83 silent misses remain among degraded photos (60 after retake).
- **Ground truth.** Open Food Facts tags are crowd-sourced, may include ‘may contain’ and miss some allergens (6.4).
- **Photos.** The 104 photos are already-cropped ingredient shots with little tilt or reflection; rectification triggered on 3/104.
- **OpenCV-only share.** Only 6.7% of verdicts are decided by the classical gate alone; we do not claim OpenCV-only judgement beyond that. No user study.

8 Responsible use

Topic	What we do
Not medical advice	No verdict says a product is safe to eat. “No warning” and “warning” both end with “final check is always by a person, on the original label”. When unsure, the system asks for a human check instead of saying “none”.
Data rights	Photos: Open Food Facts contributors, CC BY-SA; data: ODbL / DbCL; package designs belong to their makers. The repository holds no images, only product-page links and a full attribution list.
Privacy	OpenCV 5 YuNet face detection (confidence 0.8) on all 104 evaluation photos found a face in 0 of 104 photos. Uploaded photos are processed in memory and discarded; logs hold only verdict kind, latency and group count.
Security and abuse	Least-privilege IAM role (logs only), no secrets in the function, concurrency cap 5, body cap 6 MB, input type check.
AI use disclosure	The code was written with an AI coding assistant. The judging path uses classical OpenCV 5 and OpenCV DNN text models only; it calls no generative model, and no model was trained. We do not present the coding assistant as an agentic workflow.
Honest reporting	All numbers come from files produced by the evaluation scripts; unfavourable results (useful abstention 0.26, glare silent misses, recall 0.87 below the 0.90 target) are reported as measured.

Reproduction: `pytest -q` (79 tests, including 7 SAM-template checks), `python -m label_guard.adapters.evaluate linux` (104-photo tables), `scripts/measure_endpoint.py` (latency). Evaluation files: `docs/eval/` in the repository.